



CSL0506
MACHINE LEARNING

PBL ON

Fraud News Detection

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1. Abstract

Fake news spreads quickly on social media and can cause misunderstanding and confusion. To help prevent this, we built a machine learning model that can identify whether a news article is real or fake. We used TF-IDF to convert text into numbers and a Passive-Aggressive Classifier to predict the label. The model gives good accuracy and works well for basic fake news detection.

2. Introduction

2.1 Background

News is easy to access today, but fake news also spreads widely. It is difficult for people to manually check every article. Machine learning can help by automatically learning patterns from text.

2.2 Problem Statement

Manual fake news detection is slow and not practical for large amounts of news. Some current methods do not understand writing patterns well. A simple but effective model is needed to classify text as real or fake.

2.3 Objectives

- Clean and prepare text data.
- Convert text into numeric form using TF-IDF.
- Train a Passive-Aggressive Classifier.
- Evaluate accuracy, precision, recall, and F1-score.

3. Methodology

3.1 Dataset Description

- File: fake_or_real_news.csv
- Columns: -
 - a. text (news content)
 - b. label (REAL or FAKE)
- Binary classification dataset.

3.2 Data Preprocessing

- Removed punctuation
- Lowercased text
- Removed stop words
- Applied TF-IDF vectorization

3.3 Feature Engineering: TF-IDF

TF-IDF shows how important a word is in a document. It reduces the weight of common words and gives higher weight to important words.

3.4 Classification Model: Passive-Aggressive Classifier

The PAC model updates itself only when it makes a mistake. It is fast and good for large text datasets.

4. Implementation and Experimental Setup

- **Environment:** Python (Visual Studio Code).
 - **Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn.
 - **Data Split:** 80% Training, 20% Testing (Stratified to maintain class balance).
 - **Model:** Passive Aggressive Classifier (max_iter=50)
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5. Results and Analysis

5.1 Model Accuracy Comparison

The model reached high accuracy (around 92%–96%).

5.2 Classification Report

The classification report includes:

- **Precision** – reliability of positive predictions
- **Recall** – ability to find all positive samples
- **F1-score** – harmonic mean of precision and recall

The PAC model showed strong balanced performance across both classes.

5.3 Confusion Matrix

A simple confusion matrix:

- True Positives: correctly predicted fake news
- True Negatives: correctly predicted real news
- False Positives: real news predicted as fake
- False Negatives: fake news predicted as real

```
array([[587,  41],  
       [ 44, 595]], dtype=int64)
```

6. Discussion

The combination of TF-IDF and the Passive-Aggressive Classifier demonstrates high effectiveness for fake news detection. TF-IDF

captures important textual patterns, while PAC efficiently learns linear boundaries in high-dimensional sparse space. The model's robustness and fast convergence make it suitable for real-time or large-scale fake news detection systems.

Key reasons for strong performance:

1. **TF-IDF** reduces noise by assigning proper weight to informative words.
2. **PAC** handles huge feature spaces better than many classical algorithms like Naive Bayes or KNN.
3. Minimal preprocessing is needed, making the system computationally efficient.

7. Conclusion

This project successfully implemented a machine learning–based fake news detection system using TF-IDF vectorization and a Passive-Aggressive Classifier. The model achieved high accuracy and produced reliable results across standard evaluation metrics. The approach is lightweight, efficient, and easily deployable for large-scale monitoring of online news content. This serves as a solid baseline for future extensions using deep learning or hybrid ensemble models.

8. References

- Scikit-learn Documentation — Text Classification and Linear Models
- “Fake News Detection Using Machine Learning” — Various academic sources
- Dataset: fake_or_real_news.csv (user-uploaded)
- Research papers on TF-IDF and Fake News Classification

